Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[spanish]babel [T1]fontenc [utf8]inputenc pgfplots, helvet, verbatim width=10cm, compat=newest backgrounds

Esta gráfica es la segunda versión del ejemplo titulado "Espectro de Banda Base OFDM Multiportadora" https://www.overleaf.com/latex/examples/espectro-de-banda-base-ofdm-multiportadora/xhbwfnpmmhpy. A diferencia del ejemplo previamente publicado por mí, en este caso se usó la opción gnuplot [raw gnuplot] de la orden . Al comparar ambos ejemplos, se observan algunas diferencias, probablemente

por la forma en que se calculan las coordenadas en el paquete PGFPLOTS y la opción GNUPLOT que usa órdenes del paquete homónimo.

Para el trazado de las 3 curvas se usa la orden y dentro del bucle, la orden + con las opciones adecuadas para que las curvas aparezcan en colores diferentes. Otras opciones usadas se explican por sí mismas en los comentarios. cromaGoldenrod, Blue, Red

[legend style=fill=LightGrey!15, font=,anchor=north east, axis background/.style=fill=LightGrey!15, title style=text width=10cm, title=**RoyalBlue Espectro de Banda Base OFDM Multiportadora 2**, cycle list name = croma, xlabel=VioletRed $\frac{f}{N*Rs}$, ylabel=VioletRed $S(f)$, minor tick num=9, grid=major,] ıin 1, ...,3 +[no marks, line width = 1, samples=100, smooth] gnuplot [raw gnuplot] set xr[-0.1:2]; f(x) = (sin(pi*x)/(pi*x))^2; $g1(x) = sum[n = 0 : 3]f(4 * x - n); g2(x) = sum[n = 0 : 15]f(16 * x - n); g3(x) = sum[n = 0 : 63]f(64 * x - n); plotgß(x);$; N=4,N=16,N=64